YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 5 OF 10 · 40 MINUTES

Spot the pattern + Checkpoint 发现规律 + 阶段检测

A graph turns a table of numbers into a picture, so we can spot the pattern and write a conclusion that uses the data as evidence.

- I can choose between a bar chart and a line graph and plot my results accurately.
- I can write a conclusion using 'As the ___ increases, the ___ ___.'

graph 图表 | bar chart 条形图 | line graph 折线图 | axis (axes) 坐标轴 | plot 描点 | line of best fit 最佳拟合线 | pattern 规律 | conclusion 结论 | increase 增加

THE LESSON

Do Now	3	Mean of 1.40, 1.46, 1.52 s (1.46 s) and units; hand out the group envelopes.
Checkpoint	10	Alone, in silence; Chinese answers accepted; no help; collect at exactly 10 minutes.
Hook	2	Same means as a bar chart and a line graph: which tells the story better? Vote with a reason.
Learn	5	Words or groups: bar; numbers: line. IV on x-axis, DV on y-axis; crosses; line of best fit.
Do	14	Copy means (own or sample card), label axes, plot, line of best fit, conclusion frame.
Think and Reflect	6	Vote BAR or LINE on cards A to D with the frame; exit ticket and logbook line; collect sheets.

Hands-on: No practical: after the checkpoint, students plot the Week 4 spinner means (or sample data) and conclude.

Collect or assess: Checkpoint out of 12 (into tracker column Q); graph and conclusion (Rubric strand 4); exit ticket: bar or line? (bar)

SAFETY

- No practical hazards beyond normal classroom rules.
- Carry rulers and pencils safely; do not wave them near faces.

WATCH OUT

- Checkpoint overruns: collect at exactly 10 minutes and count papers against the register.
- Sample data card Mean column is blank: groups work out means, leaving out the 1.12 s anomaly at 8 cm.

Language: Frame: 'As the ___ increases, the ___ ___.' Band 1 sits with the language buddy; Chinese accepted everywhere.

Next week: Week 6 starts the ice project: ice cube rescue (Plan A) or keep it warm (Plan B), using the freezer decision.

PREP FOR ONE CLASS

Plan A: before the lesson about 40 min (23 could be done by someone else); after about 119 min (some times estimated)

Plan B: before the lesson about 40 min (23 could be done by someone else); after about 129 min (some times estimated)

THE WEEK BEFORE

- [ ] Print worksheets x24 (grid at 100%), checkpoints x25, cards A to D x1 set 13 min · reprographics
- [ ] Print and cut 24 BAR | LINE voting cards (test sheet first) or reuse a set 5 min · technician
- [ ] Print teacher copies: kit card, mark scheme, lesson plan, handover forms 5 min · teacher

THE DAY BEFORE

- [ ] Check group envelopes; note which groups need the sample data card 5 min · teacher
- [ ] Sharpen 24 pencils plus spares; spares and sharpener on teacher desk 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out kit; incoming teacher's seat at the back with plan and checkpoint 7 min · teacher

AFTER THE LESSON

- [ ] Clean-up: checkpoints counted first; logbooks to box; keep 2 spinner sets 4 min · students
- [ ] Mark checkpoints out of 12 (no half marks); enter in tracker column Q 40 min · teacher
- [ ] Read graphs, conclusions and exit tickets; note who struggled with axes 15 min · teacher
- [ ] Complete one handover note per class; book the 30-minute Week 6 walkthrough 22 min · teacher
- [ ] Strip trial: make 12 x 2 cm block, fold 5 foil troughs; build, clear 1 rig 23 min · technician
- [ ] Strip trial (Plan B): allergy check, mix salt dough, weigh 5 balls of 18 g (Plan B) 10 min · technician
- [ ] Strip trial (Plan A): freeze 5 troughs of 12 ml for 24 h+, test 3, keep 2 (Plan A) 15 min · teacher
- [ ] Strip trial (Plan B): press balls into troughs, dry a week, test 3, keep 2 (Plan B) 15 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided, grid at 100%)
- Checkpoint paper x25 (1 for incoming teacher; 1 sheet, double-sided)
- BAR | LINE voting cards x24 (3 A4 card sheets, double-sided, cut)
- Bar or line? cards A to D, 1 set (4 A5 sheets, single-sided)
- Teacher copies: kit card, mark scheme, lesson plan, handover note form

GET OR MAKE (NEW THIS LESSON)

- 30 cm rulers x24
- Pencil sharpener x1
- Kitchen foil x5 pieces, about 20 x 10 cm (strip trial)
- Template block 12 x 2 cm (wood offcut or taped rulers)
- Foil troughs x5 and plain trial strips x5 (make)
- Salt dough balls x5, 18 g (flour, salt) (Plan B)
- Kitchen scale, 1 g (Plan B)
- Measuring cylinder, 25 ml (or 20 ml syringe) x1
- Freezer space for 5 troughs, 24 h+ (Plan A)
- Dry shelf for 5 strips, one week (Plan B)
- Textbooks x6 (two stacks of three)
- Thin plastic bags x2 (Plan A)
- Carrier or freezer bag with handles x1, string
- Plastic cup x1 and marker pen (25 ml pour cup)
- Bucket or deep washing-up bowl x1
- Jug, about 2 litres x1

Have ready from earlier weeks: Class box, Logbooks x24, Group envelopes x6 (spinners, results), Role card sets x6, Sentence frame card sets x6, Science Dictionaries x24, Sample data cards x6 (Week 4), Class pencils x24, sharpened